

Known but Unreachable: A Diagnostic Evaluation of Wolof Entity Linking against Wikidata

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Abstract

Entity linking (EL) remains largely unexplored for low-resource African languages, where linking failures are generally attributed to the limited coverage of multilingual knowledge bases such as Wikidata. In this study, we show that the limited accessibility of entries that are already present in these knowledge bases constitutes a second, much less studied factor. This lack of accessibility results from the systematic mismatch between the surface forms of entities as they appear in text and the labels under which these entities are indexed in knowledge bases. To distinguish between these two factors, we introduce a diagnostic protocol applied to **WolofEL**, a benchmark dataset derived from MasakhaNER 2.0 that we manually annotated for the entity linking task and that contains 2,049 mentions, with a double-annotated sample used to estimate annotation reliability. The protocol evaluates each entity under two conditions: first using its surface form as it appears in the text, and then using its corresponding Wikidata label (oracle condition), allowing us to measure the exact cost of surface-form variation. Our results show that surface-form variation leads to a 42.72-point drop in linking accuracy for lexical retrieval and a 33.45-point drop for mGENRE, a multilingual autoregressive linker designed to generalise across language-specific surface forms, whereas genuinely missing entities account for 16.84% of all mentions and mainly correspond to local Senegalese entities. Dense retrieval with LaBSE fails to close this gap even under the oracle condition, while generative linking recovers only once the oracle label is supplied. These findings indicate that improving entity linking for low-resource African languages requires addressing not only knowledge base coverage, but also the accessibility of existing knowledge.

Keywords

Entity Linking, Wolof, Low-Resource Languages, Wikidata, Surface-Form Variation, Knowledge Base Accessibility

1. Introduction

Entity linking grounds textual mentions to entities in structured knowledge bases such as Wikidata [2], and underpins applications ranging from question answering and information retrieval to knowledge graph population and fact verification [3]. Despite substantial progress in multilingual entity linking, existing benchmarks and retrieval methods remain overwhelmingly centered on high-resource languages, leaving many African languages almost entirely unexplored [4], [5], [3]. The success of these methods, however, rests on an assumption that is rarely made explicit: that the mentions found in text can be matched to the representations under which entities are indexed in the knowledge base.

At first glance, the African language gap appears to be a simple resource issue: without annotated datasets, developing and evaluating high-performance entity linking systems becomes exponentially more difficult [6],[7]. However, low-resource languages face a second challenge that has received considerably less attention. An entity may be correctly represented in a multilingual knowledge graph and yet remain un retrievable, because the surface forms used in naturally occurring texts differ systematically from its indexed labels. Both situations, a missing entity and an unreachable one, produce the same observable outcome: a linking failure. Telling them apart requires a dedicated experimental design.

Wolof makes this phenomenon directly observable because of the diversity of its writing systems.[8]. In recent years, sustained efforts have produced linguistic resources for the language, including a Universal Dependencies treebank [9], part-of-speech annotations [10], and named entity recognition corpora [7, 11]. Yet no dedicated entity linking resource exists for the language: apart from ParaNames

NL4AI 2026: Ninth Workshop on Natural Language for Artificial Intelligence, October 6-9th, 2026, Perugia, Italy [1]

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[12], which extracts canonical Wolof name–QID pairs from Wikidata without contextual mentions or NLP decisions, entity linking remains unaddressed, as confirmed by surveys of NLP for Senegalese languages [13, 14]. More importantly, Wolof texts written in the official¹ Latin orthography [15] follow standardized native spellings that systematically diverge from the corresponding Wikidata labels.

The consequence is concrete: a Wolof journalist writing about the capital city will write *Ndakaaru*, but the entity holding its identifier in Wikidata is labeled *Dakar*. Similar divergences affect local institutions, public figures, and culturally specific entities, so that a substantial part of the knowledge relevant to Wolof text exists in the knowledge base without being reachable from the language itself.

This observation raises the following question: when entity linking fails in a low-resource language, how much of the failure is caused by missing knowledge, and how much results from the inability to retrieve knowledge that already exists? Existing benchmarks provide little means to answer it: Wikidata-based evaluation resources remain English-centric and underexploit the multilingual structure of the knowledge base [16, 3], and the contribution of surface-form mismatches to linking failure has remained unquantified.

To address this question, we introduce WolofEL², a gold standard entity linking dataset for Wolof built from the MasakhaNER 2.0 [11], together with a diagnostic protocol that disentangles the two sources of error by comparing retrieval from raw Wolof surface forms against retrieval from the corresponding Wikidata labels. Evaluating lexical retrieval, LaBSE dense retrieval, the mGENRE multilingual autoregressive linker [17], and large language models under this controlled setting, we find that surface-form mismatch alone accounts for a 42.72-point drop in In-KB accuracy for lexical retrieval and a 33.45-point drop for mGENRE, while roughly 16.84% of mentions are genuinely absent from Wikidata, concentrated among local Senegalese entities. That a generative linker explicitly designed to abstract away from language-specific surface forms suffers a comparable loss indicates that the problem is not confined to string matching. The evaluated LaBSE-based dense retrieval approach, moreover, underperforms lexical search even under oracle conditions, pointing to a representation gap that compounds the mismatch problem.

2. Related Work

Early benchmarks such as AIDA-CoNLL [18] framed entity disambiguation as a monolingual task; the field has since moved toward multilingual and cross-lingual evaluation. Mewsli-9 [4] demonstrated that multilingual evaluation at large scale is feasible, while dense bi-encoder architectures such as BLINK [5] established dense retrieval as a dominant paradigm for candidate generation and ranking. This paradigm was later extended to the multilingual setting by systems such as mGENRE [17], which autoregressively generates language-tagged entity titles over 105 languages, and BELA [19], which performs end-to-end entity linking across 97 languages. Nevertheless, these developments remain almost exclusively focused on high-resource languages: African languages are absent from standard EL benchmarks such as AIDA-CoNLL and Mewsli-9, and the few low-resource efforts conducted under the LORELEI program [6] for languages such as Tigrinya, Oromo, and Kinyarwanda relied on restricted annotation frameworks that were never publicly released. Even multilingual systems with broad language coverage are seldom evaluated on languages whose surface forms diverge systematically from knowledge base labels, so it remains unclear whether conclusions drawn from high-resource settings transfer to languages with limited digital resources and distinct writing conventions.

The scarcity of evaluation resources is only one aspect of the problem. Wikidata [2] has become the de facto knowledge base for multilingual entity linking because of its multilingual labels and collaborative structure. Systems such as OpenTapioca [20] illustrate this Wikidata-centric paradigm by performing entity linking directly over Wikidata labels and aliases. Recent surveys nevertheless

¹The official Latin orthography of Wolof was first standardized by the Senegalese government in Decree No. 71-566 of May 21, 1971, and subsequently revised by Decree No. 2005-992 of October 21, 2005, which defines the current orthographic conventions.

²<https://huggingface.co/datasets/mbaye930/WolofEntityLinking>

show that Wikidata-based entity linking resources remain strongly biased toward English despite the multilingual design of the knowledge base [16, 3].

Beyond the availability of evaluation resources, the knowledge base itself also exhibits substantial imbalances. Mbengue [21] shows that low-resource african languages remain severely underrepresented in linked open data graphs, both in the availability of labels and in the coverage of locally relevant entities. This scarcity is confirmed by ParaNames [12], the largest parallel name corpus extracted from Wikidata, which covers Wolof through canonical labels alone and reports only a marginal gain when used as a gazetteer for Wolof named entity recognition, in contrast to the larger gains observed for better-represented languages. ParaNames is the closest existing resource connecting Wolof strings to Wikidata identifiers, but it provides canonical name–QID pairs, not mentions annotated in context, and cannot mark entities absent from the knowledge base; it documents the cause of the accessibility problem without measuring its effect. Existing work has therefore largely interpreted linking failures as a consequence of incomplete knowledge base coverage. Much less attention has been paid to the complementary question of whether entities that already exist in the knowledge base are actually reachable from the surface forms used in low-resource african languages.

In parallel, dense multilingual retrieval has emerged as a promising direction for cross-lingual entity linking. Models such as LaBSE [22] project mentions and entity descriptions into a shared semantic space, reducing dependence on exact lexical matching. Recent work in the biomedical domain, such as BioELX [23], further shows that enriching the alias space with Wikidata-derived multilingual aliases can substantially improve cross-lingual retrieval for low-resource languages. Yet the behavior of these approaches under systematic surface-form variation has received little attention, and to our knowledge no study has investigated whether dense multilingual representations can compensate for native written forms that differ substantially from the labels stored in the target knowledge base.

The difficulty of reaching entities from low-resource surface forms is itself documented in the cross-lingual literature: Zhou et al. [24] separate the coverage of the underlying resources from the recall of the gold candidate, and Fu et al. [25] characterise the ways a mention can diverge from the names available to the system. We build directly on this line, turning the distinction into a measurement through a protocol that quantifies coverage and accessibility separately over the same mentions.

Wolof provides an ideal testbed for these questions, combining a documented divergence between standardized native forms and knowledge base labels, an active resource-building community, and the absence of a dedicated entity linking benchmark [13, 14].

This work differs from previous studies in two respects. First, rather than treating all linking errors as a single phenomenon, we distinguish failures caused by genuine knowledge base absence from those caused by inaccessible surface forms. Second, we introduce a controlled evaluation protocol that quantifies the contribution of each factor independently. While the distinction between missing and unreachable entities is implicit in prior low-resource work [24, 25], we are not aware of a study that isolates the two experimentally and reports the cost of surface-form variation as a single measured quantity; to our knowledge none has done so for an African language.

3. The WolofEL Dataset

3.1. Source Corpus

WolofEL is derived from MasakhaNER 2.0 [11], currently the only publicly available manually annotated named entity recognition resource for Wolof, which makes it the appropriate starting point for constructing an entity linking evaluation set. We use the validation and test splits and intentionally exclude the training split at this stage. The dataset construction involved a two-stage filtering process. First, at the sentence level, any sentence lacking a named entity mention was discarded. Second, at the mention level, DATE spans were excluded, as temporal normalization constitutes a distinct task from entity linking; consequently, only LOC, ORG, and PER mentions were retained. This process yields a final corpus of 1,045 sentences containing 2,049 entity mentions.

3.2. Corpus Statistics

Type	In-KB	NIL	Total	QIDs	NIL %
LOC	758	23	781	174	2.94
ORG	420	128	548	173	23.36
PER	526	194	720	222	26.94
Total	1,704	345	2,049	565	16.84

Table 1: WolofEL dataset statistics.

Table 1 provides the complete details. LOC mentions exhibit the lowest NIL rate (2.94%), reflecting the comprehensive indexing of Senegalese geographical entities on Wikidata. PER and ORG display substantially higher NIL rates (26.94% and 23.36%), driven by local public figures and domestic organizations absent from global knowledge graphs. Among the 1,704 linkable mentions, approximately 90% carry a surface form that diverges from the standard Wikidata label.

3.3. Annotation Procedure

Each mention is annotated with its original surface form as written in the Wolof source text, its gold Wikidata QID for linkable mentions, the corresponding standard Wikidata label and description, and an annotation status field documenting how the link was established. To maximize initial recall against surface-form and translational variation, candidate QIDs were pre-generated by querying the Wikidata search API in three languages (Wolof, French, and English). Automated queries alone proved insufficient: because of the systematic divergence described in Section 1, candidate sets were frequently noisy or empty. Every candidate was therefore manually verified then validated, and mentions with no valid candidate underwent exhaustive manual search, including cross-referencing entity descriptions across French and English, and wolof, before being assigned a NIL label. A mention is marked NIL only when this search confirms that the entity has no Wikidata entry.

3.4. Annotation Decisions

The annotation of Wolof text against a global knowledge base raised recurring dilemmas that required explicit rules; these decisions define the semantics of the dataset and are documented here so that users can interpret the labels correctly. When an individual is referenced solely by a family name, such as *Mbaye*, the mention is annotated NIL rather than linked to the generic patronym QID that Wikidata provides, since mapping multiple distinct local individuals onto a single abstract surname entry would introduce systematic bias. Abbreviated or variant person names, such as *Pep* for Pep Guardiola, are resolved to a single consistent QID by cross-referencing descriptions across the French and English pivot languages. Typographical errors inherited from the source corpus, such as *Aifrig* for *Afrig* (Africa, Q15), are linked to the correct QID whenever sentence context resolves the ambiguity, with the annotation status marked as *manual* to signal human intervention. Spans that concatenate several distinct entities as a result of upstream segmentation errors, such as *PS PDS APR*, cannot be split at the linking stage and are marked NIL. Senegalese institutional acronyms such as *SDE* or *ICS* are disambiguated through manual expansion to their full names before querying, since bare acronym queries systematically return irrelevant international entities. Historical name changes are resolved to the QID of the continuous legal entity: when a name appearing in the text has no Wikidata entry but a former or subsequent name of the same institution does, the mention is linked to that existing QID to preserve institutional continuity. Finally, placeholder names and initials used in press articles to anonymize individuals are marked NIL, as they intentionally denote unidentifiable persons.

3.5. Annotation Reliability

A second native Wolof speaker independently re-annotated a stratified sample of 180 mentions, balanced across entity types and across linked and NIL decisions, working blind to the original labels and applying the guidelines above. Agreement reaches a Cohen’s κ of 0.74 on the link/NIL decision, which is substantial, with 100% exact-match agreement on the QID (116/116) for mentions both annotators chose to link.

The 20 disagreements fall into three recurring types: bare patronyms subject to the family-name NIL rule (*Keyta*, *Faal*, *Silmani*), acronyms with competing local and international expansions (*ICS*, *AFP*), and locally significant or homonymous entities whose Wikidata coverage is ambiguous (*Belli Torodo*, *Missirah*). Disagreement thus concentrates at the coverage/accessibility boundary: once both annotators agree that a mention is linkable, they select the same QID without exception. All disagreements were subsequently adjudicated to a single gold label.

4. Evaluation Protocol

Using WolofEL, we design a controlled evaluation whose objective is not merely to compare entity linking systems, but to isolate the respective contributions of knowledge base coverage and surface-form accessibility to linking failures.

4.1. Diagnostic Protocol

The protocol rests on two conditions applied to every mention. In condition A, the system receives the surface form exactly as it appears in the Wolof text. In condition B, the surface form is replaced by the standard Wikidata label of the gold entity; mentions annotated as NIL receive a NIL prediction directly, since no standard label exists to substitute. Because this label is the form under which the entity is indexed, condition B guarantees lexical reachability, allowing the impact of surface-form variation to be isolated independently of knowledge base coverage, candidate generation, or model capacity.

Condition B serves as an oracle rather than a competing system: it relies on gold information unavailable at inference time and is intended solely to estimate the performance recoverable through perfect surface-form normalization. The protocol is therefore diagnostic rather than comparative, using the evaluated systems as instruments for measuring the effect of surface-form variation rather than as the object of the study.

Systems that perform their own candidate generation or entity generation are evaluated under both conditions, since the oracle substitution is meaningful only when the system is free to retrieve or generate from the substituted query; this applies to the two retrieval baselines and to mGENRE. The LLMs operate over a fixed candidate list generated by the lexical baseline and are therefore evaluated under condition A only; their role in the protocol is to test whether generative models can partially compensate for surface-form variation through contextual reasoning over candidates.

4.2. Compared Systems

The systems are chosen to cover four paradigms rather than to maximize the number of models. Lexical retrieval represents the direct string-matching approach; dense retrieval represents a strong embedding-based retrieval paradigm; autoregressive generation represents the dominant multilingual entity linking architecture, which produces an entity identifier directly instead of ranking retrieved candidates; and LLM-based linking represents contextual disambiguation over candidates, including a model specialized for Wolof.

The lexical baseline queries the public Wikidata Search API independently in three languages (Wolof, French, and English), retrieving up to 15 results per language. All experiments were performed in May 2026 using the same API configuration, which was kept identical under both evaluation conditions, so that the comparison between conditions is unaffected by the evolving state of Wikidata. The retrieved

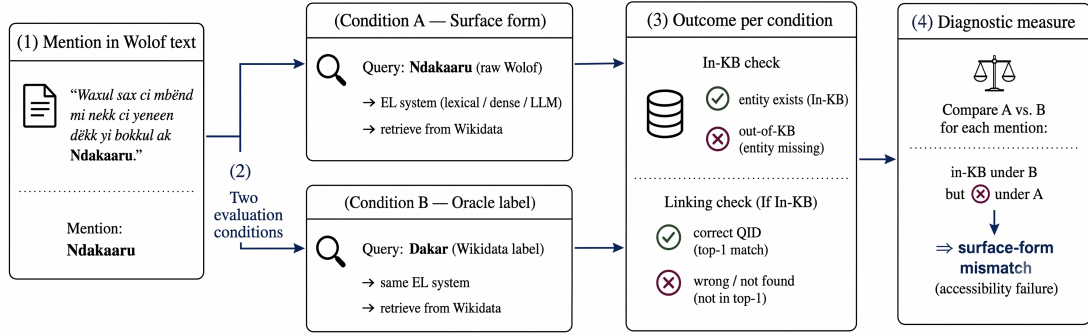


Figure 1: Diagnostic evaluation pipeline. Each mention is evaluated under two conditions: the raw Wolof surface form (A) and the corresponding Wikidata label (B, oracle). Comparing the outcomes disentangles failures due to missing knowledge (out-of-KB in both conditions) from failures due to surface-form mismatch (in-KB under B but wrong or not found under A).

candidate lists are released together with the code, so that the reported numbers can be recomputed exactly. The returned items are deduplicated by QID, yielding a single ranked candidate list of up to 45 entries. Under condition A, this list is built from the raw Wolof surface form; under condition B, the standard Wikidata label replaces the surface form as the query string. The top-ranked candidate after deduplication is selected as the system prediction, while the ranking metrics (MRR, Candidate Recall) are computed over the full ordered list prior to this top-1 selection. The baseline predicts NIL when the API returns an empty candidate list.

The evaluated LaBSE-based dense retrieval approach [22] is used in a retrieve-and-rank pipeline. Candidate generation follows the same lexical retrieval procedure as the baseline, while LaBSE is exclusively used for reranking, by cosine similarity between the query embedding and each candidate’s label and description embedding. A mention is predicted NIL when the best cosine score falls below a threshold of 0.40, set by grid search over [0.25, 0.55] with a step of 0.05 on the validation portion of the data; this is the only tuned hyperparameter, and the In-KB accuracy results are independent of it, since In-KB accuracy is computed only over linkable mentions. The encoding strategy differs between conditions. Under condition A, the full sentence context is encoded to compensate for surface-form noise; under condition B, only the mention string is encoded, since the standardized label already provides an unambiguous lexical representation. This asymmetry is empirically motivated. As Table 2 shows, the full sentence is best under condition A and the mention alone is best under condition B. Each condition therefore uses its best encoding in the main results. The accessibility gap does not depend on this choice: held fixed, the A→B gap stays large under both encodings.

Encoding (fixed A/B)	Cond. A	Cond. B	Δ
Mention only	10.39	43.90	+33.51
Full sentence	21.36	36.27	+14.91

Table 2: LaBSE In-KB accuracy (%) with a single encoding strategy fixed across both conditions. The A→B gap stays large under either encoding.

We also evaluate mGENRE [17], an autoregressive linker that generates a language-tagged entity title instead of ranking retrieved candidates. We use the released checkpoint with constrained beam search (5 beams) over the official 105-language title trie, so that every generated string resolves to a Wikidata identifier, and marginalise predictions across languages following the reference implementation. The mention is marked with the standard [START]/[END] delimiters, the surface form being replaced by the gold label under condition B. Two properties matter here: constrained decoding always yields a valid title, so mGENRE never predicts NIL under condition A, and its candidate space comes from the trie alone, never from the Wikidata Search API.

The LLM-based systems are GPT-4o mini³ and Gemini 2.5 Flash⁴ as general-purpose commercial models, and Oolel-v0.1 [26], a 7.6B open-weights model based on the Qwen-2.5 architecture. Each model receives the context sentence, the target mention, the candidate list retrieved by the lexical baseline, and three static few-shot examples (one instance each of LOC, ORG, and PER, of which two are linkable and one is NIL). The model must select the correct QID or output NIL: it does so implicitly when the candidate list is empty, or explicitly when none of the retrieved candidates match the mention in context. The complete prompt templates and exemplars are given in Appendix A.

4.3. Metrics

In-KB Accuracy is the primary metric, as it directly measures whether a linkable entity was reached; it is computed over linkable mentions only. NIL detection is evaluated with precision, recall, and F1, since recognizing that an entity is absent is a structurally different capability from ranking candidates. An overall micro F1 across all mentions summarizes both. For retrieval-based systems, Candidate Recall measures the proportion of gold entities present in the retrieved candidate list, isolating retrieval failures from ranking failures, and Mean Reciprocal Rank measures how highly the correct candidate is ranked when present. These two metrics are not reported for the LLMs, which operate over a fixed candidate list, nor for mGENRE, which generates a single title rather than ranking a candidate set. Confidence intervals are estimated by paired bootstrap resampling over mentions (10,000 resamples), each mention being drawn jointly under both conditions so that the interval on the $A \rightarrow B$ difference accounts for their dependence.

5. Results

Table 3, 4, and 5 summarize the performance of the evaluated systems.

System	In-KB	F1	NIL F1	Rec. Cands	MRR
Lexical A (Raw)	46.89	49.59	41.97	57.45	0.50
LaBSE A	21.36	29.48	41.24	57.69	0.33
Lexical B (Oracle) [†]	89.61	91.36	99.86 [†]	99.12	0.94
LaBSE B (Oracle) [†]	43.90	53.34	99.57 [†]	99.24	0.63

Table 3: Results (%) for retrieval-based systems under conditions A and B. [†] Under Condition B, NIL mentions are assigned directly (oracle-assisted).

System	In-KB Acc.	Overall F1	NIL F1
<i>Condition A (raw Wolof surface form)</i>			
GPT-4o-mini	50.82	56.61	42.98
Oolel-v0.1 (7.6B)	54.75	56.08	45.00
Gemini 2.5 Flash	56.10	61.93	46.10
mGENRE	48.36	40.21	0.00
<i>Condition B (oracle)</i>			
mGENRE [†]	81.81	84.87	99.71 [†]

Table 4: Results (%) for generative systems. The LLMs operate few-shot over the lexical candidate list and are evaluated under condition A only; mGENRE performs its own constrained generation and is evaluated under both conditions. Bold indicates the best condition A score per column.

³<https://openai.com/fr-FR/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>

⁴<https://ai.google.dev/gemini-api/docs/models/gemini-2.5-flash?hl=fr>

System	Type	In-KB	F1	NIL F1	Rec. Cands	MRR
Lexical A	LOC	56.99	56.98	12.26	65.04	0.60
	ORG	41.43	41.42	40.00	63.81	0.50
	PER	36.69	47.78	54.22	41.44	0.38
LaBSE A	LOC	12.14	13.83	11.72	65.04	0.26
	ORG	27.62	31.75	39.32	64.52	0.43
	PER	29.66	44.72	55.70	41.63	0.34
mGENRE A	LOC	68.21	66.20	0.00	–	–
	ORG	40.48	31.02	0.00	–	–
	PER	26.05	19.03	0.00	–	–
Lexical B	LOC	85.36	85.79	100.00	99.87	0.92
	ORG	87.86	90.69	99.61	97.14	0.92
	PER	97.15	97.92	100.00	99.62	0.98
LaBSE B	LOC	25.46	27.66	97.87	99.87	0.49
	ORG	54.29	64.96	99.61	97.14	0.70
	PER	62.17	72.36	99.74	100.00	0.76
mGENRE B	LOC	84.30	84.76	100.00	–	–
	ORG	70.00	77.01	99.22	–	–
	PER	87.64	90.97	100.00	–	–
GPT-4o-mini	LOC	55.28	56.34	12.39	–	–
	ORG	56.43	60.04	51.54	–	–
	PER	39.92	54.31	53.87	–	–
Gemini 2.5	LOC	63.46	64.53	15.03	–	–
	ORG	61.90	66.24	56.13	–	–
	PER	40.87	55.83	54.60	–	–
Oolel-v0.1	LOC	63.59	63.38	18.18	–	–
	ORG	59.52	55.11	39.85	–	–
	PER	38.21	48.89	54.32	–	–

Table 5: Per-type results (%) under conditions A and B. Candidate Recall and MRR are omitted for the LLMs, which reuse the lexical baseline’s candidate list, and for mGENRE, which generates a single title rather than ranking candidates. As in Table 3, Condition B NIL F1 scores are oracle-assisted

5.1. Retrieval Performance

Under condition A, the lexical baseline reaches 46.89% In-KB Accuracy with a Candidate Recall of 57.45%. The LaBSE reranker, operating over the same candidate lists, obtains 21.36% In-KB Accuracy. Under condition B, the lexical baseline rises to 89.61% In-KB Accuracy and 99.12% Candidate Recall, while LaBSE reaches 43.90%. The gap between conditions amounts to 42.72 points for the lexical baseline and 22.54 points for LaBSE. Both gaps are statistically significant: paired bootstrap resampling over mentions yields 95% confidence intervals of [40.26, 45.13] and [19.66, 25.47] respectively, neither of which includes zero. To verify that validation-tuning does not inflate threshold-dependent metrics, we evaluated LaBSE on the test split alone. Results remain almost identical (In-KB accuracy 21.16% vs. 21.36% overall in ConditionA; 42.77% vs. 43.90% in ConditionB), preserving the 21.61-point gap. We thus report full-corpus figures in the main tables. Across both conditions, the LaBSE reranker consistently performs below the lexical baseline operating on the same candidate lists. NIL detection follows the same trend: under condition B, both systems achieve near-perfect NIL F1 (99.86 and 99.57), whereas under condition A performance drops sharply to 41.97 and 41.24. These condition B NIL scores should be read with care, since NIL decisions are made by construction under the oracle rather than by the system itself.

Table 5 breaks down these results by entity type. The accessibility gap between conditions A and B is

present for all three types but differs substantially in magnitude. On the lexical baseline, In-KB accuracy rises from 56.99% to 85.36% for LOC, from 41.43% to 87.86% for ORG, and from 36.69% to 97.15% for PER, corresponding to gaps of 28.4, 46.4, and 60.5 points respectively. The LLM-based systems show the same relative ordering across types, with their strongest In-KB accuracy on LOC and ORG and their weakest on PER. Taken together, these results indicate that surface-form variation is a major source of entity linking failures in Wolof. We analyze these findings in more detail in Section 7.

5.2. Robustness to within-sentence correlation.

Because several mentions can occur in the same sentence, the mention-level bootstrap may underestimate uncertainty. We therefore repeat the analysis with sentence-level clustered resampling, drawing 1,045 sentences with replacement rather than individual mentions. As Table 6 shows, the clustered intervals are only slightly wider than the mention-level ones about 0.3 pp on the lexical gap and every gap remains significant at $p < 0.05$. Within-sentence dependence therefore has little effect on our uncertainty estimates.

System	Bootstrap	CI Cond. A	CI Cond. B	CI $\Delta_{A \rightarrow B}$
Lexical	Mention	[44.54, 49.30]	[88.15, 91.08]	[40.26, 45.13]*
	Clustered	[44.21, 49.53]	[88.05, 91.09]	[39.91, 45.45]*
LaBSE	Mention	[19.37, 23.36]	[41.61, 46.24]	[19.66, 25.47]*
	Clustered	[19.21, 23.50]	[41.33, 46.52]	[19.31, 25.80]*
mGENRE	Mention	[45.95, 50.76]	[79.93, 83.63]	[30.93, 35.92]*
	Clustered	[45.63, 51.07]	[79.77, 83.82]	[30.70, 36.30]*

Table 6: Mention-level vs. sentence-level clustered bootstrap 95% CIs for In-KB Accuracy.

5.3. Generative Entity Linking

Under condition A, mGENRE reaches 48.36% In-KB accuracy (95% CI [45.95, 50.76]), statistically indistinguishable from the lexical baseline (46.89%, 95% CI [44.54, 49.30]) despite being purpose-built for multilingual entity linking, and far below the 91.9% reported for Wolof in the original evaluation [17]. Under condition B it rises to 81.81%, an accessibility gap of 33.45 points (95% CI [30.93, 35.92]).

mGENRE does cover Wolof, but on a very thin support of 1,503 pages and 7,257 hyperlinks, and its reported accuracy is measured on Wikipedia hyperlinks, where mentions already coincide with indexed titles. On the press forms of WolofEL the same model loses roughly a third of its accuracy relative to the oracle, so surface-form variation penalises a generative linker built to abstract away from language-specific forms almost as heavily as it penalises lexical retrieval (42.72 points). Its NIL F1 of zero under condition A is a structural consequence of constrained decoding rather than a ranking failure, and its condition B NIL score reflects the oracle protocol rather than system behaviour.

Table 5 reproduces the ordering observed for retrieval, the gap being smallest for LOC (16.1 points) and largest for PER (61.6 points). Finding it again in a system that performs no string matching points to a property of Wolof naming practice. The aggregate similarity with the lexical baseline under condition A also hides a trade-off: mGENRE gains 11.2 points on LOC, where phonetic adaptation preserves enough of the international form to be reconstructed, but loses 10.6 points on PER.

To check that frequent entities do not dominate the mention-level figures, we recompute the accessibility gap over the 565 unique QIDs. Under condition A the lexical baseline resolves at least one mention correctly for 41.8% of unique entities (macro-averaged accuracy 35.4%), against 90.1% under condition B (macro 89.5%) an entity-level gap of 48.3 pp in coverage and 54.0 pp in macro accuracy. For mGENRE, entity coverage rises from 48.7% to 78.9%. The gap thus holds at the entity level as well as the mention level, so it does not stem from a few frequent entities.

6. Ablation Study

A natural objection is that the accessibility gap could be closed by simple orthographic normalization. We test this by transforming the Wolof surface form before retrieval under three schemes of increasing strength. The first strips diacritics and maps non-ASCII characters to their base letters ($\hat{a} \rightarrow a$, $\acute{o} \rightarrow o$, $\tilde{n} \rightarrow n$, $\eta \rightarrow n$). The second applies rule-based phonetic transliteration toward the French digraphs commonly used for Wolof sounds ($\tilde{n} \rightarrow gn$, $\eta \rightarrow ng$, $x \rightarrow kh$) and reduces long vowels to a single character ($aa \rightarrow a$). The third combines both. In each case the normalized mention is sent to the same retrieval pipeline as the raw baseline. Table 7 shows that none of these schemes improves retrieval. Every scheme lowers In-KB accuracy below the raw baseline of 46.89%, down to 42.31% for the most aggressive one, and reduces Candidate Recall from 57.45% to 54.17%. Far from approaching the oracle bound of 89.61%, normalization moves performance in the opposite direction. Two effects explain this. Normalization destroys the matches that already worked on the original form, since Wikidata indexes some entities under their exact Wolof or international spelling; and mechanical transliteration produces forms that match neither the standardized Wolof nor the official French label, adding noise rather than bridging the gap. The accessibility gap is therefore not an artifact of diacritics or of a simple character mapping. It is a structural mismatch that lightweight normalization cannot resolve, which is precisely what makes it a research problem rather than a preprocessing detail.

Normalization (cond. A)	In-KB	F1	NIL F1	R.can	MRR
None (raw)	46.89	49.59	41.97	57.45	0.50
Diacritic stripping	45.83	48.71	41.97	57.34	0.50
Transliteration	43.54	46.41	41.02	54.34	0.47
Combined	42.31	45.39	41.02	54.17	0.47
Oracle (cond. B) [†]	89.61	91.36	99.86 [†]	99.12	0.94

Table 7: Effect of lightweight orthographic normalization on the lexical baseline under condition A, with the oracle (condition B) as upper bound. No scheme improves retrieval; all reduce In-KB accuracy below the raw baseline.

7. Discussion

Four findings emerge from this study. First, the major obstacle to Wolof entity linking is not only incomplete knowledge base coverage, but also the limited accessibility of entities that are already present in the knowledge base. Second, the diagnostic protocol shows that surface-form variation accounts for the majority of linking failures, costing the lexical baseline 42.72 accuracy points and removing the gold entity from the candidate list in over 40% of cases. Third, the accessibility gap is not an artefact of string matching: mGENRE, a multilingual autoregressive linker that generates entity titles rather than matching strings, loses 33.45 points between the two conditions, of the same order as the 42.72 points lost by the lexical baseline. Fourth, the evaluated LaBSE-based approach consistently underperforms the lexical baseline, even under the oracle condition where surface-form variation is eliminated. Importantly, this accessibility gap cannot primarily be explained by limitations of the retrieval API itself. Both conditions use exactly the same Wikidata Search API configuration, including the same languages, ranking mechanism, and candidate limits. Yet Candidate Recall increases from 57.45% under the raw Wolof surface forms to 99.12% under the oracle labels. This indicates that the dominant source of retrieval failure is the query representation rather than API truncation or ranking bias. mGENRE settles the point: it never queries the search API, generating titles under a trie constraint that covers the full multilingual title space, and it still shows a 33.45-point gap. Surviving a complete change of candidate-generation mechanism, the gap must originate in the surface forms themselves.

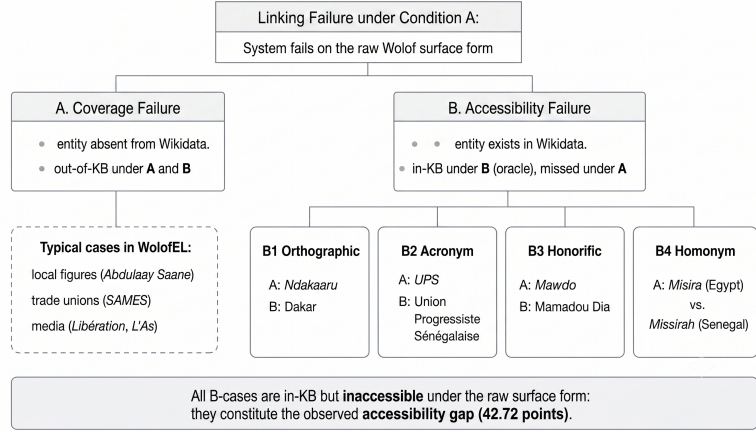


Figure 2: Typology of data-side linking failures under condition A, decomposed into coverage failures (entity absent from Wikidata) and accessibility failures (entity in Wikidata but unreachable from its Wolof surface form), with the four subtypes observed in WolofEL. Model-side failures (Section 7) are not depicted.

7.1. Error Analysis

The failure mechanisms behind these findings fall into two families that mirror the central distinction of this work, along with a third family located in the models themselves.

Accessibility failures concern entities that exist in Wikidata but cannot be reached from their Wolof written form. The most frequent mechanism is surface-form divergence. Wolof orthography systematically adapts foreign names and uses characters such as $\tilde{n}$, η , $\acute{o}$, $\grave{a}$, and $\tilde{a}$ [27] that are absent from Wikidata labels, which follow French or English conventions. Countries and geographical locations are adapted phonetically or designated by local historical names: Ivory Coast appears as *Kótdiwaar* (Q1008), Saint-Louis as *Ndar* (Q273779), and Thiès as *Cees* (Q600693). Political figures undergo even more radical phonetic and structural transformations: Antoine Félix Abdoulaye Diome appears as *Ántuwaan Féliks Joom* (Q101437800), Aly Ngouille Ndiaye as *Aali Nguy Njaay* (Q98730622), and Abdoulaye Daouda Diallo as *Abdulaay Daawuda Jállo* (Q98730700). These mentions fail because string matching is character-sensitive and Wikidata provides native Wolof aliases for almost none of these entities; this single mechanism explains most of the Candidate Recall collapse from 99.12% to 57.45%.

A second mechanism is phonetic and cultural adaptation specific to Wolof news discourse: surnames are transcribed phonetically (*Dia* becomes *Ja*, *Dias* becomes *Jaas*, *Diagne* becomes *Jaañ*) and public figures are designated by single-token surnames or honorifics: *Mawdo* (meaning the Elder) designates historical Prime Minister Mamadou Dia (Q739956), and *Jomaay* alone designates President Bassirou Diomaye Faye (Q124382437). These forms fail because no lexical or semantic path connects an isolated honorific or truncated surname to its encyclopedic identity: querying *Ja* or *Mawdo* yields generic items or irrelevant international individuals.

The type-wise gaps quantify this ordering: person names show the largest accessibility gap (60.5 points) and locations the smallest (28.4 points). Toponyms often keep a recognizable form across writing conventions, while person names concentrate phonetic transcription, truncation, and honorific substitution, none of which offer a lexical path to the Wikidata label. That the LLMs remain closest to the lexical baseline precisely on PER confirms that person-name accessibility is not recovered by contextual reasoning over candidates.

A third mechanism is acronym ambiguity: short strings such as *SDE*, *ICS*, or *UPS* return globally known entities rather than the intended Senegalese organizations, with *UPS* retrieving United Parcel Service rather than the historical Union Progressiste Sénégalaise (Q3550454), because global popularity dominates API ranking. Homonym collisions operate similarly: the mention *Misira* designates Egypt (Q79) in Wolof text but collides with the Senegalese locality Missirah (Q3316893), and systems default to the wrong entity.

Coverage failures concern the mentions whose entities are absent from Wikidata. A manual audit of

the 345 NIL mentions shows 329 genuine absences (95.36%). The remaining 16 (0.78% of the benchmark) are upstream annotation artefacts: spans concatenated by MasakhaNER 2.0 (*PS PDS APR*, merging three political parties), anonymised initials from crime reporting, and isolated first names identifying no specific individual. The genuine coverage gap is thus 16.06%, close enough to the raw 16.84% to leave our conclusions unchanged. Genuine absences are concentrated among local public figures such as municipal leaders (*Abdulaay Saane*), regional trade unions (*SAMES*), and domestic media outlets (*Libération, L'As*). These mentions fail in a specific way: rather than correctly outputting NIL, systems hallucinate links to international entities with similar acronyms or partial name overlap, which explains the low NIL precision of all systems under Condition A.

A third family of failures is located in the models rather than in the data. Even under oracle labels, LaBSE fails on more than half of the linkable mentions and degrades the ranking produced by lexical search. Two mechanical factors are likely to contribute: encoding a short mention against longer candidate descriptions distorts cosine similarity, and the NIL threshold calibrated on Condition A does not transfer to Condition B. The uneven descriptive coverage of Wikidata aggravates both [21]. In WolofEL, 25 of the 565 gold QIDs carry no French or English description, and all of them are local Senegalese entities, so their candidate embeddings rest on a short label alone.

Overall, improving entity linking for Wolof requires addressing accessibility and coverage jointly: in our audit, only 29% of the 565 gold QIDs have a Wolof label and 3% a Wolof alias, leaving retrieval little native form to match against and limiting alias-centric approaches (e.g., OpenTapioca [20]).

7.2. Limitations

This study is limited to Wolof and to a single knowledge base (Wikidata) and source corpus (MasakhaNER 2.0), so coverage/accessibility patterns may differ in other languages, genres, or resources. While annotation quality is supported by double annotation on 180 mentions (Section 3), a full second pass would further strengthen reliability. Finally, we rely on the Wikidata Search API and evaluate a limited model panel; future work should test alternative indexing/ranking strategies and broader end-to-end systems (e.g., BELA [19]).

8. Conclusion

WolofEL helps answer a key question in low-resource entity linking: are entities missing, or merely unreachable? Most linkable entities are already in Wikidata, but surface-form variation drives a large accessibility gap (42.72 points for lexical search; 33.45 points for mGENRE).

We release WolofEL (2,049 mentions with explicit NIL) and a diagnostic protocol to separate coverage from accessibility, described in Section 3 and 4. Future work should focus on retrieval robust to spelling variation and on enriching Wikidata with Wolof labels and aliases.

Declaration on Generative AI

During the preparation of this work, the author used DeepL for translation and ChatGPT for language and formatting assistance (e.g., grammar, clarity, and LaTeX editing). After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the final content of the publication.

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A. Prompt Templates for LLM-based Entity Linking

This appendix details the exact instructions, formatting guidelines, and few-shot exemplars utilized in our LLM candidate selection evaluations. All models were queried with temperature 0.0 to ensure deterministic outputs.

A.1. Few-Shot Exemplars Selection

To align with the retrieve-and-rank framework, three static exemplars were selected from our development guidelines. Each exemplar is structured to cover different aspects of the entity linking task:

System Prompt:

You are an expert Entity Linking system. Your task is to link a named entity mention in a Wolof sentence to its correct Wikidata QID from a list of candidates. You must respond with ONLY the QID (e.g. Q1025) or 'NIL' if no candidate corresponds.

User Message Template:

Here are some examples of the task:

Example 1:

Sentence in Wolof:

"Chelsea door na Manchester United."

Entity to link:

- Mention: "Chelsea"
- NER Type: ORG

Wikidata Candidates:

- 1. QID: Q9616 | Label: Chelsea F.C. | Description: club de football anglais
- 2. QID: Q170119 | Label: Chelsea | Description: quartier de Londres

Decision:

Q9616

Example 2:

Sentence in Wolof:

"Jeynaba Taraawore amoon na lu war a mat ñaar-fukki at."

Entity to link:

- Mention: "Jeynaba Taraawore"
- NER Type: PER

Wikidata Candidates:

(Aucun candidat trouvé sur Wikidata)

Decision:

NIL

Example 3:

Sentence in Wolof:

"Dakar mooy péyub Senegaal."

Entity to link:

- Mention: "Senegaal"
- NER Type: LOC

Wikidata Candidates:

- 1. QID: Q1041 | Label: Sénégal | Description: pays d'Afrique de l'Ouest
- 2. QID: Q3511306 | Label: Fleuve Sénégal | Description: fleuve de l'Afrique de l'Ouest

Decision:

Q1041

Now, perform the task for the following input:

Sentence in Wolof:

"{sentence}"

Entity to link:

- Mention: "{mention}"
- NER Type: {ner_type}

Wikidata Candidates:

{candidates_list}

Instructions:

1. Select the correct Wikidata QID for this entity based on the Wolof context.
2. If the entity is not in the candidate list, or is not in Wikidata, respond with 'NIL'.
3. Reply with ONLY the QID (e.g., Q1025) or 'NIL'. Do not include any explanations.

Decision:

B. Online Resources

The code and data for this work are available via

- [GitHub \(Code WolofEL\)](#).
- [WolofEL Dataset Hugging Face](#)